

Laboratory Exercise 1: Flowcharts and Pseudocode

Learning Outcome:

- Implement flowcharts and pseudocode to solve a given problem.

Table 1: Operator Symbols

Statement	Description	Example
+ - * / %	Add, subtract, multiply, divide, modulo (i.e. get remainder)	x + 7
=	Assigns a value to a variable	x = 7
< <= > >= == !=	Comparison operators (Lesser than, Lesser than or equal, greater than, greater than or equal, equal, not equal)	x<10
&&	Relational operators (And , Or)	x=0 y=1 x && y

Table 2: Flowchart Symbols

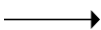
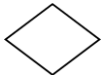
Shape	Name	Description
	Flow Line	The arrow would direct the process, it must begin from one symbol and then terminate to another
	Terminal	This would contain the start or end of the program.
	Process	Used to show that a process is performed. E.g. z = x + y
	Decision	Commonly answerable by yes or no, true or false. The arrow that would indicate the direction must always be labeled if it is the true path or the false path.
	Manual Input	Any input operation done manually
	Preparation	This shows the operation that is required to declare and initialize a data prior to a process or condition.
	Display	Indicates a process step where information is displayed to a person (e.g., PC user, machine operator).

Table 3: Pseudocode Statements

Statement	Description	Example
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display	Outputs item on screen	display 4
input	Ask for user input	input x
if ... else end	Indicates a branching operation	x = 5 if x < 5 display "Less than 5" else if x>5 display "Greater than 5" end
while end	Indicates an repetitive operation	x = 5 while x > 0 display x x = x - 1 end

Exercise Details

Using the operator symbols in Table 1, the standard flowchart symbols in Table 2 and pseudocode statements Table 3 to provide the flowchart and pseudocode to solve the indicated problems. Note: That the operator symbols can be used both on the flowchart and the pseudocode.

Problems:

1. Compute simple grading system using the formula:

$$\text{Grade} = (\text{TE} * 0.5 + \text{Q} * 0.3 + \text{A} * 0.2).$$

Where:

TE is the term exam input

Q is the quiz input

A is the assignment input

Then display the literal equivalent of the grade as shown below:

Grade	Literal Equivalent
91 – 100	A
81-90	B
71-80	C
61-70	D
51-60	E
Below 50 but non-zero	F
Above 100 or below 0	No Such Grade Exist

2. Perform division of a $\mathbb{N} > 0$ using repeated subtraction as described by the process flow below

Example:

Dividend: 10

Divisor:3

Sub-process results:

10 – 3 = 7, Partial Quotient: 1

7 – 3 = 4, Partial Quotient: 2

4 – 3 = 1, Partial Quotient: 3

Quotient: 3, Remainder: 1

Take note that the dividend must always be greater than or equal to the divisor